(Please write your Enrollment Number) **Enrollment No._____**

MID-TERM EXAMINATION

(Course Name: B. Tech.) (Semester: 6)

(March 2025) OFF LINE mode

Subject Code: BAI 306	**Subject: Digital Image Processing**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1	**Attempt all the following:**	**(2.5*4=10)**	**CO Mapping**
	a) Describe the process of image sampling and quantization. Explain with the necessary diagrams.	2.5	CO1
	b) Define the following pixel relationships: Neighbors, Connectivity, Adjacency, and Distance Measures.	2.5	CO1
	c) What are the key differences between spatial and frequency domain filtering in terms of their techniques, applications, and effects on image enhancement?	2.5	CO2
	d) What is the Fourier Transform, and how is it useful in image processing? Also, explain the Butterworth Low-pass filter.	2.5	CO2
	UNIT I		**CO Mapping**
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) Explain the fundamental steps involved in a digital image processing system. Give a diagram or flowchart to explain.	5	CO1
	b) An image segment is given below. Let V be the set of gray-level values used to define connectivity in the image. Compute the shortest De, D4, D8, and Dm distance between pixels 'p' and 'q' for: (i) V = {1, 4} (ii) V = {1, 5} Given Image Segment: $\begin{bmatrix} 1(p) & 2 & 1 & 5 & 3 \\ 4 & 1 & 2 & 4 & 5 \\ 3 & 5 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 2 & 5(q) \end{bmatrix}$	5	CO1
	c) Given the 5×5 grayscale image matrix. Assume that the intensity values range from 0 to 255 wherever required.: $I = \begin{bmatrix} 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 7 & 8 & 9 \\ 6 & 7 & 8 & 9 & 10 \end{bmatrix}$ Perform the following point operations on the matrix: 1. Apply contrast stretching r_{min}=2, r_{max}=10, s_{min}=0 and s_{max}=50. 2. Apply Power-Law (Gamma) Transformation with c = 1 and γ = 0.5. 3. Apply Thresholding using a threshold value 6.	5	CO1

UNIT II			CO Mapping
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) Discuss different smoothing and sharpening filters in both the spatial domain and Frequency Domain with supporting equations and an example of a 3x3 image matrix.	5	CO2
	b) Perform Histogram Equalization on the given 3×3 grayscale image matrix, where pixel intensities range from 0 to 7. Given Image Matrix: $$I = \begin{bmatrix} 2 & 3 & 3 \\ 4 & 5 & 6 \\ 6 & 7 & 7 \end{bmatrix}$$ Find the equalized image for the above matrix.	5	CO2
	c) Given the 5×5 grayscale image matrix: $$I = \begin{bmatrix} 10 & 20 & 30 & 40 & 50 \\ 20 & 30 & 40 & 50 & 60 \\ 30 & 40 & 50 & 60 & 70 \\ 40 & 50 & 60 & 70 & 80 \\ 50 & 60 & 70 & 80 & 90 \end{bmatrix}$$ Apply the following filters on the above matrix at position (2,2) and compute the new intensity value. Assume row and column index starting at (0,0) 1. Apply the 3×3 averaging filter for the center pixel value 50 at (position (2,2)). 2. Apply the 3×3 Gaussian Blur filter for the center pixel value 50 at (position (2,2)). 3. Apply the Sobel Filters Sx and Sy separately on the image matrix and compute the gradient magnitudes and edge direction for the center pixel value 50 at (position (2,2)).	5	CO2